

Minimax Regret Bounds in Stochastic and Adversarial Multi-Armed Bandits

with Variance-Reduction Extensions for EXP3

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Abstract—We study minimax regret rates for the K -armed multi-armed bandit (MAB) problem under both stochastic and adversarial reward models. We review the interaction protocol and regret notion, provide a detailed discussion of why minimax lower bounds scale as $\Omega(\sqrt{nK})$, and survey representative algorithms: UCB1 (stochastic), EXP3 (adversarial), and minimax-optimal policies MOSS (stochastic) and INF (adversarial) which achieve $\Theta(\sqrt{nK})$ worst-case regret (up to constants). Motivated by estimator variance in EXP3, we add two project-level extensions: (i) adaptive implicit exploration (regularizing importance weights via denominators $p_{i,t} + \gamma_t$), and (ii) baseline subtraction (a control-variate technique) which preserves unbiasedness and provably reduces conditional variance by centering rewards. We include self-contained derivations for stability properties and variance calculations, along with proof sketches and intuition to make the document a stand-alone reference for a course project.

Index Terms—multi-armed bandits, regret minimization, minimax bounds, EXP3, UCB, MOSS, INF, implicit exploration, variance reduction

I. INTRODUCTION

Multi-armed bandits (MAB) are a minimal model of online learning with partial feedback. A learner repeatedly selects one of K actions (arms) and observes only the reward of the chosen arm. This captures scenarios such as adaptive clinical trials, online advertisement selection, recommendation systems, and resource allocation, where counterfactual outcomes for unchosen actions are not observed.

A central challenge is the *exploration–exploitation tradeoff*:

- **Exploration:** try different arms to learn which are good.
- **Exploitation:** focus on the best-looking arm to accumulate reward.

Bandit algorithms aim to balance these objectives so that the learner performs nearly as well as the best fixed arm in hindsight. This is formalized through *regret*.

A key theme of this report is the *minimax* viewpoint: rather than focusing only on easy instances, we ask what performance is possible in the worst case over environments. In both stochastic and adversarial bandits, the minimax regret rate scales as $\Theta(\sqrt{nK})$ in horizon n and number of arms K , but achieving this rate requires different techniques. For adversarial environments, exponential-weight algorithms such as EXP3 provide robust guarantees, while INF removes an additional $\sqrt{\log K}$ factor via implicit normalization. For stochastic environments, optimism-based strategies such as UCB1 provide strong instance-dependent bounds but are not

minimax-optimal; MOSS modifies the exploration bonus to achieve minimax-optimal $\sqrt{nK}$ behavior.

Finally, we include two variance-reduction extensions to EXP3. These are not meant to improve the minimax rate by themselves, but to provide algorithmic ideas that can improve stability and empirical behavior: (i) *implicit exploration* by regularizing importance weights, and (ii) *baseline subtraction* as a control variate.

II. PROBLEM SETUP

A. Interaction Protocol

Fix $K \geq 2$ arms and horizon n . At each round $t = 1, \dots, n$:

- 1) The learner chooses a distribution $p_t = (p_{1,t}, \dots, p_{K,t})$ over arms.
- 2) The environment specifies a reward vector $g_t = (g_{1,t}, \dots, g_{K,t}) \in [0, 1]^K$.
- 3) The learner draws $I_t \sim p_t$ and observes only $g_{I_t,t}$ (partial feedback).

B. Models: Stochastic vs. Adversarial

Stochastic bandits. Each arm i has an unknown distribution ν_i on $[0, 1]$. When arm i is pulled, a reward is drawn i.i.d. from ν_i with mean μ_i . The randomness comes from the environment itself, and past observations are informative about future rewards.

Adversarial bandits. The environment may choose g_t arbitrarily (possibly as a function of the learner’s past actions), subject only to $g_{i,t} \in [0, 1]$. Here, past rewards need not predict future ones; hence algorithms rely on worst-case robustness rather than statistical inference.

C. Regret

Definition 1 (Regret to the best fixed arm). *The (expected) regret with respect to the best fixed arm is*

$$R_n = \max_{i \in [K]} \mathbb{E} \left[\sum_{t=1}^n (g_{i,t} - g_{I_t,t}) \right]. \quad (1)$$

The goal is to design a policy with $R_n = o(n)$, meaning the average regret R_n/n vanishes as $n \rightarrow \infty$.

D. Why Compare to the Best Fixed Arm?

In adversarial bandits, competing with stronger benchmarks (e.g., a changing comparator) is generally impossible without additional structure. The best fixed arm is the standard baseline that remains achievable: it captures the idea that the learner should perform nearly as well as the single best action in hindsight, even if the environment is chosen adversarially.

III. MINIMAX LOWER BOUND: WHY $\sqrt{nK}$ IS UNAVOIDABLE

We sketch why $\Omega(\sqrt{nK})$ regret is information-theoretically unavoidable. The key idea is to construct environments that are hard to distinguish from the learner's partial observations.

A. A Two-Instance "Confusion" Argument (High Level)

Consider two stochastic instances that are identical except for one arm. Intuitively:

- If the learner does not explore an arm enough, that arm might secretly be optimal, causing regret.
- If the learner explores too much, it wastes pulls on suboptimal arms, also causing regret.

A typical construction chooses gaps Δ small enough that distinguishing the two instances requires many samples; optimizing Δ yields $\sqrt{nK}$.

B. Role of KL Divergence

Let P and Q be the distributions over the learner's observed history under two environments. A change-of-measure inequality (one form relates total variation to KL divergence) implies that if $\text{KL}(P\|Q)$ is small, then no algorithm can reliably tell which environment generated the data. In bandits, $\text{KL}(P\|Q)$ scales with the number of times the "differing" arm is pulled; hence at least one of the instances forces large regret.

Remark 1. *This lower bound philosophy applies to both stochastic and adversarial settings (with different formalizations). It provides the benchmark that motivates minimax-optimal algorithms such as INF and MOSS.*

IV. PRIOR WORK AND MAIN ALGORITHMS

This section expands the "previous works" discussion, emphasizing the algorithmic mechanisms and what their bounds mean.

A. Adversarial Bandits: EXP3

1) *Algorithmic idea:* EXP3 (Exponential-weight algorithm for Exploration and Exploitation) is designed for adversarial rewards. It maintains weights $w_{i,t} > 0$ and samples arms according to probabilities

$$p_{i,t} = \frac{w_{i,t}}{\sum_{j=1}^K w_{j,t}}. \quad (2)$$

Because only $g_{I_t,t}$ is observed, EXP3 uses an *importance-weighted* (inverse propensity) estimator:

$$\hat{g}_{i,t} = \frac{g_{I_t,t} \mathbf{1}\{I_t = i\}}{p_{i,t}}. \quad (3)$$

This estimator is unbiased for each $g_{i,t}$ but can have high variance if $p_{i,t}$ is small. Weights are updated via

$$w_{i,t+1} = w_{i,t} \exp(\eta \hat{g}_{i,t}), \quad (4)$$

where η is a learning rate.

2) *Regret guarantee (interpretation):* With suitable tuning, EXP3 guarantees

$$R_n \leq \mathcal{O}(\sqrt{nK \log K}).$$

This matches the minimax lower bound up to a $\sqrt{\log K}$ factor. The extra factor arises from worst-case concentration control and the need to ensure sufficient exploration, typically enforced via mixing or implicit terms.

B. Adversarial Bandits: INF (Implicitly Normalized Forecaster)

1) *Motivation:* INF targets the gap in EXP3 by changing how probabilities are formed. Instead of exponential weights with a standard normalization, INF uses a function ψ and chooses a normalizer C_t such that

$$p_{i,t} = \psi(\tilde{G}_{i,t} - C_t), \quad \sum_{i=1}^K p_{i,t} = 1, \quad (5)$$

where $\tilde{G}_{i,t}$ are estimated cumulative gains. The normalization is implicit: C_t is defined as the unique value satisfying the summation constraint.

2) *Why this helps:* The mapping via ψ can be chosen to control how quickly probabilities collapse. This changes the stability/regularization behavior of the forecaster and enables a sharper analysis.

3) *Regret guarantee:* With a suitable ψ , INF achieves

$$R_n \leq \mathcal{O}(\sqrt{nK}),$$

closing the $\sqrt{\log K}$ gap in the adversarial minimax rate (up to constants).

C. Stochastic Bandits: UCB1

1) *Optimism in the face of uncertainty:* In stochastic bandits, the learner can leverage concentration of empirical means. UCB1 assigns each arm an index:

$$\text{UCB}_i(t) = \hat{\mu}_i(t) + \sqrt{\frac{2 \log t}{T_i(t)}}, \quad (6)$$

where $T_i(t)$ is the number of pulls of arm i before time t and $\hat{\mu}_i(t)$ is the empirical mean of observed rewards from arm i .

2) *Intuition and proof sketch:* A suboptimal arm i (gap $\Delta_i = \mu^* - \mu_i$) is pulled only if either:

- its empirical mean is unusually high (over-optimism), or
- the optimal arm's empirical mean is unusually low (under-optimism).

Both events are controlled by Hoeffding-type concentration, giving $T_i(n) = \mathcal{O}(\log n / \Delta_i^2)$ and hence

$$R_n = \sum_{i: \Delta_i > 0} \Delta_i \mathbb{E}[T_i(n)] = \mathcal{O}\left(\sum_{i: \Delta_i > 0} \frac{\log n}{\Delta_i}\right).$$

TABLE I
REPRESENTATIVE REGRET GUARANTEES (ORDERS, IGNORING
CONSTANTS).

Algorithm	Setting	Regret Order
EXP3	Adversarial	$\sqrt{nK \log K}$
INF	Adversarial	$\sqrt{nK}$
UCB1	Stochastic	$\sum_{i: \Delta_i > 0} \frac{\log n}{\Delta_i}$
UCB1 (worst-case)	Stochastic	$\sqrt{nK \log n}$
MOSS	Stochastic	$\sqrt{nK}$

3) *Distribution-free limitation*: In worst-case (distribution-free) terms, UCB1 suffers an additional $\sqrt{\log n}$ factor:

$$R_n = \Theta(\sqrt{nK \log n}),$$

which motivates MOSS.

D. Stochastic Bandits: MOSS

1) *Key modification*: MOSS modifies the confidence bonus by using $\log(n/(Ks))$ instead of $\log t$:

$$B_{i,s} = \hat{X}_{i,s} + \sqrt{\frac{\max\{\log(n/(Ks)), 0\}}{s}}, \quad B_{i,0} = +\infty. \quad (7)$$

The learner plays the arm with the largest current index $B_{i,T_i(t-1)}$.

2) *Why it improves worst-case behavior*: The $\log(n/(Ks))$ structure effectively calibrates exploration to the horizon and the number of arms. It prevents overly-large bonuses that can accumulate in distribution-free analysis and is tuned to match the minimax scaling.

3) *Regret guarantee*: MOSS achieves

$$\sup R_n \leq C\sqrt{nK},$$

which matches the minimax lower bound up to constants.

E. Summary Table

V. EXTENSION I: ADAPTIVE IMPLICIT EXPLORATION FOR EXP3

The standard EXP3 estimator $\hat{g}_{i,t} = g_{I_t,t} \mathbf{1}\{I_t = i\}/p_{i,t}$ can be unstable when $p_{i,t}$ is small, due to the $1/p_{i,t}$ factor. We regularize denominators by adding $\gamma_t > 0$:

$$\tilde{g}_{i,t} = \frac{g_{I_t,t} \mathbf{1}\{I_t = i\}}{p_{i,t} + \gamma_t}. \quad (8)$$

This is biased but variance-controlled.

A. Algorithm: A-IX-EXP3

B. Stability Lemmas and Interpretation

Let $\mathcal{F}_{t-1}$ denote the history up to time $t-1$. The point of the following lemmas is to formalize why the estimator is more stable than $1/p$ importance weighting.

Lemma 1 (Uniform boundedness). *For all i, t , $0 \leq \tilde{g}_{i,t} \leq 1/\gamma_t$.*

Proof. Since $g_{I_t,t} \in [0, 1]$ and $p_{i,t} + \gamma_t \geq \gamma_t > 0$, we have $\tilde{g}_{i,t} \leq 1/\gamma_t$. Nonnegativity is immediate. $\square$

Algorithm 1 A-IX-EXP3 (Adaptive Implicit Exploration EXP3, gain form)

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1: Input: learning rates  $\{\eta_t\}_{t=1}^n$ , IX schedule  $\{\gamma_t\}_{t=1}^n$  with  $\gamma_t > 0$ 
2: Initialize  $w_{i,1} = 1$  for all  $i \in [K]$ 
3: for  $t = 1$  to  $n$  do
4:    $p_{i,t} \leftarrow \frac{w_{i,t}}{\sum_{j=1}^K w_{j,t}} \quad \forall i$ 
5:   Draw  $I_t \sim p_t$ , observe  $g_{I_t,t} \in [0, 1]$ 
6:   for  $i = 1$  to  $K$  do
7:      $\tilde{g}_{i,t} \leftarrow \frac{g_{I_t,t} \mathbf{1}\{I_t = i\}}{p_{i,t} + \gamma_t}$ 
8:      $w_{i,t+1} \leftarrow w_{i,t} \exp(\eta_t \tilde{g}_{i,t})$ 
9:   end for
10: end for

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Lemma 2 (Second-moment control). *Let $\tilde{g}_{I_t,t} = \sum_{i=1}^K \mathbf{1}\{I_t = i\} \tilde{g}_{i,t}$. Then*

$$\mathbb{E}[\tilde{g}_{I_t,t}^2 \mid \mathcal{F}_{t-1}] \leq \frac{1}{\gamma_t}. \quad (9)$$

Proof. Condition on $\mathcal{F}_{t-1}$. Then

$$\mathbb{E}[\tilde{g}_{I_t,t}^2] = \sum_{i=1}^K p_{i,t} \left(\frac{g_{i,t}}{p_{i,t} + \gamma_t} \right)^2 \leq \sum_{i=1}^K \frac{p_{i,t}}{(p_{i,t} + \gamma_t)^2}.$$

Using $\frac{p}{(p+\gamma)^2} \leq \frac{1}{\gamma(p+\gamma)}$,

$$\sum_{i=1}^K \frac{p_{i,t}}{(p_{i,t} + \gamma_t)^2} \leq \frac{1}{\gamma_t} \sum_{i=1}^K \frac{p_{i,t}}{p_{i,t} + \gamma_t} \leq \frac{1}{\gamma_t}.$$

$\square$

Lemma 3 (Controlled pessimism (bias)). *For each arm i ,*

$$\mathbb{E}[\tilde{g}_{i,t} \mid \mathcal{F}_{t-1}] = \frac{p_{i,t}}{p_{i,t} + \gamma_t} g_{i,t} \leq g_{i,t}, \quad (10)$$

and $0 \leq g_{i,t} - \mathbb{E}[\tilde{g}_{i,t} \mid \mathcal{F}_{t-1}] \leq \gamma_t$.

Proof. Since $\Pr(I_t = i \mid \mathcal{F}_{t-1}) = p_{i,t}$,

$$\mathbb{E}[\tilde{g}_{i,t} \mid \mathcal{F}_{t-1}] = p_{i,t} \cdot \frac{g_{i,t}}{p_{i,t} + \gamma_t}.$$

Then $g_{i,t} - \mathbb{E}[\tilde{g}_{i,t}] = g_{i,t} \frac{\gamma_t}{p_{i,t} + \gamma_t} \leq \gamma_t$ using $g_{i,t} \leq 1$. $\square$

Remark 2 (How to read these lemmas). *The classic EXP3 estimate can be as large as $1/p_{i,t}$, which may explode if some arm's probability becomes tiny. Adding γ_t prevents this. This does not automatically improve minimax regret without a careful analysis of the induced bias, but it provides a mathematically grounded stabilization mechanism.*

VI. EXTENSION II: BASELINE SUBTRACTION (CONTROL VARIATES) FOR EXP3

Baseline subtraction is a variance-reduction technique that preserves unbiasedness. Let b_t be any scalar chosen based only on past information (i.e., b_t is $\mathcal{F}_{t-1}$ -measurable). Define:

$$\hat{g}_{i,t}^{(b)} := \frac{(g_{I_t,t} - b_t) \mathbf{1}\{I_t = i\}}{p_{i,t}} + b_t. \quad (11)$$

This is a control variate: we subtract a baseline inside the importance-weighted term and then add it back to keep the mean unchanged.

A. Unbiasedness and Variance Calculations

Lemma 4 (Unbiasedness). *For every arm i and time t ,*

$$\mathbb{E}[\hat{g}_{i,t}^{(b)} \mid \mathcal{F}_{t-1}] = g_{i,t}. \quad (12)$$

Proof. Condition on $\mathcal{F}_{t-1}$. Since $\Pr(I_t = i \mid \mathcal{F}_{t-1}) = p_{i,t}$,

$$\mathbb{E}\left[\frac{(g_{I_t,t} - b_t)\mathbf{1}\{I_t = i\}}{p_{i,t}} \mid \mathcal{F}_{t-1}\right] = \frac{1}{p_{i,t}} \cdot p_{i,t}(g_{i,t} - b_t) = g_{i,t} - b_t.$$

Adding b_t yields $\mathbb{E}[\hat{g}_{i,t}^{(b)} \mid \mathcal{F}_{t-1}] = g_{i,t}$. $\square$

Lemma 5 (Conditional variance (closed form)). *For every arm i and time t ,*

$$\text{Var}(\hat{g}_{i,t}^{(b)} \mid \mathcal{F}_{t-1}) = \left(\frac{1}{p_{i,t}} - 1\right) (g_{i,t} - b_t)^2. \quad (13)$$

Proof. Let $p := p_{i,t}$, $g := g_{i,t}$, $b := b_t$. The estimator takes two values: if $I_t = i$ (prob. p), $\hat{g}_{i,t}^{(b)} = b + \frac{g-b}{p}$; otherwise it equals b . Since $\mathbb{E}[\hat{g}_{i,t}^{(b)} \mid \mathcal{F}_{t-1}] = g$, we compute

$$\hat{g}_{i,t}^{(b)} - g = \begin{cases} (g - b) \left(\frac{1}{p} - 1\right), & I_t = i, \\ -(g - b), & I_t \neq i. \end{cases}$$

Thus

$$\begin{aligned} \text{Var} &= p(g - b)^2 \left(\frac{1}{p} - 1\right)^2 + (1 - p)(g - b)^2 \\ &= (g - b)^2 \left[\frac{(1 - p)^2}{p} + (1 - p)\right] \\ &= (g - b)^2 \left(\frac{1}{p} - 1\right). \end{aligned}$$

Remark 3 (Comparison to standard importance weighting). *The standard estimator $\hat{g}_{i,t} = g_{I_t,t}\mathbf{1}\{I_t = i\}/p_{i,t}$ satisfies*

$$\text{Var}(\hat{g}_{i,t} \mid \mathcal{F}_{t-1}) = \left(\frac{1}{p_{i,t}} - 1\right) g_{i,t}^2.$$

Baseline subtraction replaces $\hat{g}_{i,t}^2$ by $(g_{i,t} - b_t)^2$, so it reduces variance when b_t tracks typical reward levels.

Lemma 6 (Conditional second moment). *For every arm i and time t ,*

$$\mathbb{E}\left[\left(\hat{g}_{i,t}^{(b)}\right)^2 \mid \mathcal{F}_{t-1}\right] = g_{i,t}^2 + \left(\frac{1}{p_{i,t}} - 1\right) (g_{i,t} - b_t)^2. \quad (14)$$

Proof. Use $\mathbb{E}[X^2] = \text{Var}(X) + (\mathbb{E}X)^2$ together with the previous two lemmas. $\square$

Algorithm 2 EXP3 with Baseline Subtraction (gain form)

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1: Input: learning rate  $\eta > 0$ 
2: Initialize  $w_{i,1} = 1$  for all  $i \in [K]$ , set  $b_1 = 0$ 
3: for  $t = 1$  to  $n$  do
4:    $p_{i,t} \leftarrow \frac{w_{i,t}}{\sum_{j=1}^K w_{j,t}} \quad \forall i$ 
5:   Choose baseline  $b_t$  using  $\mathcal{F}_{t-1}$  (e.g., running average)
6:   Draw  $I_t \sim p_t$ , observe  $g_{I_t,t} \in [0, 1]$ 
7:   for  $i = 1$  to  $K$  do
8:      $\hat{g}_{i,t}^{(b)} \leftarrow \frac{(g_{I_t,t} - b_t)\mathbf{1}\{I_t = i\}}{p_{i,t}} + b_t$ 
9:      $w_{i,t+1} \leftarrow w_{i,t} \exp(\eta \hat{g}_{i,t}^{(b)})$ 
10:   end for
11: end for

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B. Optimal Scalar Baseline for a Variance Proxy

A natural proxy for variance in EXP3-style analyses is the squared magnitude of the importance-weighted centered term on the sampled arm:

$$J(b) := \mathbb{E}\left[\left(\frac{g_{I_t,t} - b}{p_{I_t,t}}\right)^2 \mid \mathcal{F}_{t-1}\right] = \sum_{i=1}^K \frac{(g_{i,t} - b)^2}{p_{i,t}}.$$

Differentiating and setting to zero gives the minimizer

$$b_t^* = \frac{\sum_{i=1}^K \frac{g_{i,t}}{p_{i,t}}}{\sum_{i=1}^K \frac{1}{p_{i,t}}}. \quad (15)$$

In bandit feedback, b_t^* is not directly computable, but this expression clarifies what an “ideal” baseline would look like: it emphasizes arms with small probabilities (since they contribute most to variance).

C. Implementable Baselines

Two simple $\mathcal{F}_{t-1}$ -measurable choices are:

$$b_t = \frac{1}{t-1} \sum_{s=1}^{t-1} g_{I_s,s}, \quad t \geq 2, \quad b_1 := 0, \quad (16)$$

$$b_t = (1 - \alpha)b_{t-1} + \alpha g_{I_{t-1},t-1}, \quad \alpha \in (0, 1). \quad (17)$$

$\square$ These keep unbiasedness and can reduce variance when rewards have a relatively stable average scale.

D. EXP3 with Baseline Subtraction (Algorithm)

VII. CONCLUSION AND DISCUSSION

We reviewed minimax regret rates for stochastic and adversarial multi-armed bandits, and expanded the discussion of classical algorithms and why their guarantees differ. In particular, MOSS and INF achieve the minimax-optimal $\Theta(\sqrt{nK})$ regret rate in the stochastic and adversarial settings, respectively, while UCB1 and EXP3 highlight key paradigms (optimism and exponential weights) that are foundational in bandit theory.

Motivated by estimator variance in EXP3, we proposed two project-level extensions. Adaptive implicit exploration regularizes inverse-propensity weighting and yields clean boundedness/second-moment controls. Baseline subtraction preserves unbiasedness and provably reduces variance by centering rewards. These analyses are self-contained and suitable

for inclusion in a course report; future work could connect them to tighter regret analyses and empirical evaluation.

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